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IFAC PapersOnLine 50-1 (2017) 5424–5429

Virtual Synchronous Generator Strategy for VSC-MTDC and the Probabilistic Small Signal Stability Analysis

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Abstract: In this paper, a new virtual synchronous generator (VSG) strategy based on the V^2-P droop control strategy is proposed for the voltage source converter based multi-terminal high voltage direct current (VSC-MTDC) system. The VSG strategy enables the inverter stations to damp the power oscillation and provide frequency support for the AC system. Virtual inertia coefficient is obtained to couple the DC voltage and the angular frequency. To provide the oscillation damping and frequency support, the inverter stations could adjust the output power by the adjustment of the DC reference voltage. Due to the characteristic of the V^2-P droop method, there is no requirement for the communication between inverter stations. The Monte Carlo method is used to test the probabilistic small signal stability of the VSG strategy. The performance of the proposed controller is demonstrated by a hybrid AC/DC system which is modeled in DIgSILENT/PowerFactory.

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Keywords: VSG, VSC-MTDC, Monte Carlo method, probabilistic small signal stability.

1. INTRODUCTION

Offshore wind farms (OWFs) develop rapidly in recent years. The voltage source converter based multi-terminal high voltage direct current (VSC-MTDC) technology is considered to be the most appropriate method to connect the OWFs and onshore power grid. The most critical control object of the VSC-MTDC system is to maintain stable DC voltage [1], [2]. There are several mature control strategies for the VSC-MTDC system, such as voltage margin method, voltage droop method and so on [3], [4]. Most of the traditional control strategies only focus on the stability of the DC grid, while the state of the AC system is ignored. However, VSC-stations have no inherent inertia. An increasing amount of renewable energy and VSC-MTDC system may lead to a negative result that the stability of the hybrid AC/DC system will be reduced [5].

To overcome the defects of the traditional control strategies of the VSC-MTDC system, many researchers have made great efforts to develop innovative methodss. A communication-free control strategy has been proposed in [6] and [7], which could enable the inverter stations to adjust the output power according to the state of the AC system. What's more, wind turbines store a large quantity of kinetic energy which could be used to provide frequency support. However, drawing too much kinetic energy from the rotor may lead to turbines stalling. Power margin method of the wind generators are also utilized to provide the primary frequency regulation [8] and [9], which could not make full use of wind energy. In [10] and [11], a frequency droop controller is investigated. The AC grid frequency is used as the feedback signal to change the power output instructions of inverter stations. The frequency deviation could be minimized by this method and the burden of the power support is shared by all inverter stations. Among these control strategies, there is a simplified linear relationship between the frequency deviation and the DC voltage variation, which may result in an inaccurate adjustment of the power allocation ratio between the inverter stations.

A new technology called 'virtual synchronous generator' (VSG) is proposed in recent years. This method make the converters behave like a synchronous generator which could contribute inertia for the power system. The grid frequency and voltage could be coupled by using the VSG strategy. The power output of the inverter stations is changing with the state of the AC system [12] and [13]. The VSG strategy is mostly used in the distributed generators (DGs), such as photovoltaic, wind turbine and energy storage [14]. The capacity of grid-side voltage source converter (GSVSC) station is large enough to have noticeable effect on the stability of the system. The VSG technique is suitable to be applied on the VSC-MTDC system.

Because of the fluctuation of wind speed, the power output of wind turbines is inconstant. The load demand and the operation state of transmission tie-lines are also stochastic. Probabilistic small signal stability analysis (PSSSA) is often performed to evaluate the stability of the hybrid power system. The most accurate method is the Monte Carlo simulation [15].

In this paper, a novel VSG strategy based on the V^2-P control is proposed. The controller enables the GSVSCs to provide frequency support and oscillation damping. The DC reference voltage is changed autonomously in a narrow range to mimic the action of the valve of prime mover. Thus, the output power will be redistributed among the GSVSCs according to

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10.1016/j.ifacol.2017.08.1077

the state of the AC system. The Monte Carlo simulation is utilized to test the performance of the VSG strategy. The test system is modelled in DIgSILENT.

The rest of the papers are organized as followings: Part 2 presents the new VSG strategy based on the V^2-P droop method. Part 3 introduces the implication of the Monte Carlo method on the PSSSA. Part 4 shows the performance of the VSG strategy and the result of PSSSA.

2. SYNCHRONVERTER CONTROL STRATEGY

The power balance in the GSVSC is similar to synchronous generator, which can be explained by Fig. 1.

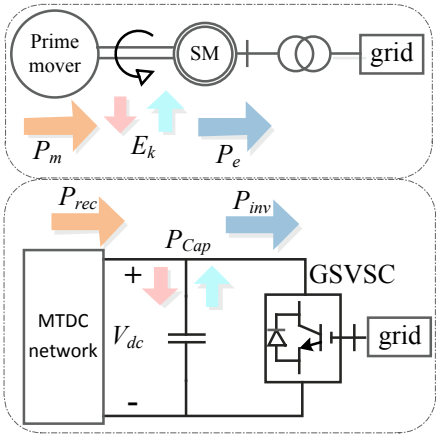


Fig. 1 The comparison of GSVSC and synchronous generator

The imbalance power of AC grid will lead to frequency deviation, which could be explained by swing equation.

$$2H\frac{d\omega_*}{dt} = P_{m*} - P_{e*} - D(\omega_* - \omega_{0*}) \tag{1}$$

where '*' represents the per unit value; ω is the angular frequency and ω_0 represent the rated angular frequency; P_m and P_e are the prime mover power and the electromagnetic power, respectively; H is the inertia constant; D represents the damping factor.

The mismatch between P_m and P_e which is caused by the change of load demand or the disturbance on the prime mover will change the rotor speed. The kinetic energy stored in the rotor could compensate partial of the imbalance power. With the deceleration (or acceleration) of the rotor speed, the governor of the generator will take a sequence of actions to adjust the position of valve of steam turbine. Thus the power output of prime mover will be adjusted until the system reaches a new equilibrium point.

There is a similar feature for the GSVSC. The power injected by DC grid can be thought as the prime mover power and the power output of the GSVSC can be considered as the electromagnetic power. In consideration of the fluctuation of wind power, the DC voltage will change instantly as long as the GSVSC did not timely adjust its reference power. The energy stored in the DC capacitors will compensate partial of the imbalance power, which could be explain by the following equation

$$\frac{CV_{dc}}{S_{vsc}}\frac{dV_{dc}}{dt} = P_{rec*} - P_{inv*} \tag{2}$$

where the S_{vsc} is the rated power capability of VSC-station; P_{rec} represents the rectifier power input to VSC-MTDC system; P_{inv} is the GSVSC power output to AC grid; C represents the capacitance of DC capacitors; V_{dc} is DC voltage of MTDC system.

According to equation (2), the V^2-P control method can be derived as

$$\Delta P = K_{droop}\Delta V_{dc}^2 \tag{3}$$

where K_{droop} is the droop coefficient; ΔV^2_{dc} is the DC voltage deviation and the ΔP is the additional power signal for GSVSC. The K_{droop} determines the power allocation ratio of the VSC-MTDC system. However, K_{droop} will keep constant in a relatively short period until the transmission system operator (TSO) change it. Thus, traditional V^2-P control method is unable to provide oscillation damping or fast frequency support.

Equation (3) indicates that GSVSC will get more power by the adjustment of the DC reference voltage. With this characteristic, the DC reference voltage would change with the frequency deviation in order to provide frequency support, which is just like the adjustment of prime mover's valve. Based on equation (1) and (2), the relationship between the DC voltage and angular frequency can be derived as

$$2H_{vsc}\frac{d\omega_*}{dt} = \frac{CV_{dc}}{S_{vsc}}\frac{dV_{dc}}{dt} \tag{4}$$

Integrating both sides of (4) yields the following equation:

$$\Delta V_{dc}^2 = \frac{4H_{vsc}S_{vsc}}{C}\Delta\omega \tag{5}$$

where $\Delta\omega$ represents the deviation of angular frequency in AC grid; H_{vsc} represents the virtual inertia constant. The physical meaning of H_{vsc} is the time required for the capacitor to reach the rated voltage from zero. The value of H_{vsc} is determined by

$$H_{vsc} = \frac{CV_{dcn}^2}{2S_{vsc}} \tag{6}$$

where V_{dcn} represents the rated voltage of GSVSC.

Considering that the regulation of the valve of steam turbine has a certain delay, the first order process should be taken into account.

$$\Delta V_{dc_ref}^2 = \frac{1}{1+sT}\Delta V_{dc}^2 \tag{7}$$

where T is the time constant. The typical range of T is from 0.1s to 0.4s.

Based on the above analysis, the virtual swing equation could be derived as

$$2H_{vsc}\frac{d\omega_*}{dt} = P_{ref0*} + K[V_{dc}^2 - (V_{dc_ref}^2 + \Delta V_{dc_ref}^2)] - P_{inv*} - D_{vsc}(\omega - \omega_0) \tag{8}$$

where $\Delta V^2_{dc_ref}$ is the correction of DC reference voltage, which can be obtained by equation (5) and (7).

Equation (8) indicates that any imbalance power (whatever is in AC side or DC side), the GSVSC will adjust its DC reference voltage to compensate the mismatch power. Thus, the VSG strategy is able to damp oscillation and provide frequency support for the AC system. What's more, there is no demand for the communication in the hybrid system. The control block diagram of VSG strategy is sketched in Fig 2.

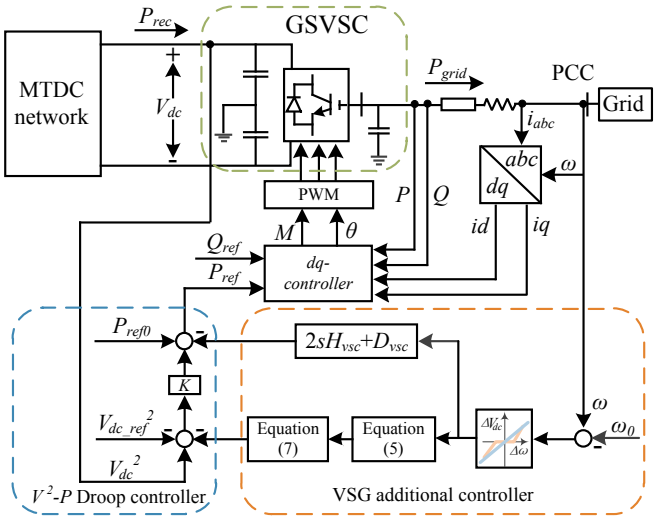


Fig. 2 The control block diagram of VSG strategy

3. PSSSA FOR the VSC-MTDC SYSTEM

The Monte Caro method can provide the most accurate evaluation of the system stability by a large number of random scenarios. Accumulation of the SSSA for all of the random scenarios constitutes the distribution of the dominant oscillation modes.

3.1 Distribution of random variables

(1) Random wind speed

Two-parameter Weibull distribution is used to describe the random wind speed:

$$f(v;c,k)=\frac{k}{c}\left(\frac{v}{c}\right)^{k-1}e^{-\left(\frac{v}{c}\right)^{k}} \tag{9}$$

where v is the wind speed; c represents the scale parameter, which reflects the mean wind speed; k represent the shape parameter, which reflects the characteristic of the distribution of wind speed. In this paper, c=11m/s, k=2.06. The random wind speed can be obtained by

$$v=-c\sqrt[k]{\ln(r_w+1)} \tag{10}$$

where r_w is a random number which generated by the uniform distribution.

(2) Random load demand

The load demand generally meets the normal distribution, which can be obtained by the Box-Muller method.

$$P_m=P_{m0}(1+\sigma\%\cdot\sqrt{-2\log r_1}\cdot\cos 2\pi r_2) \tag{11}$$

where P_{m0} and P_m represent the mean and random load value respectively; r_1 and r_2 are the random number generated by the uniform distribution.

(3) Random operation state of transmission tie-lines

The operation state of transmission tie-lines are mutual-independent. A transmission line will be cut off according to the uniform distribution between [0, L] (L represents the total number of transmission lines) in each random scenario.

3.2 The process of PSSSA and the stability index

The Weak Law of Large numbers and the Central Limit Theorem are the foundations of the Monte Caro simulation. The random operation modes of the power system are independent of each other. According to the SSSA, the power system is instable when any one of the system eigenvalues' real part σ is positive. The notation R_i is used to represent the stability of a random scenario of the power system

$$R_i=\begin{cases}1, & \alpha_i>0\\ 0, & \alpha_i<0\end{cases} \tag{12}$$

The random variable R_i meets the 0-1 distribution. When R_i equals to 1, it means that the power system is instable. Thus, the instability of the system can be measured by the expected value. A more detailed introduction of the application of the Monte Carlo method to power system can be found in [15].

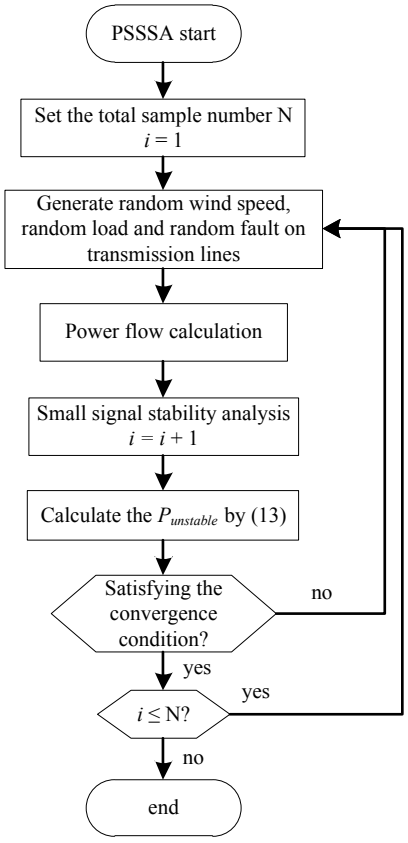


Fig. 3 The procedure of PSSSA

According to the Monte Caro simulation, a lot of results are obtained to evaluate the system stability. So long as the real

part of eigenvalue is positive, the power system is considered to be instable. The probability of instability can be defined as the following equation

$$P_{unstab} = \overline{R}_n = \frac{R_1 + \cdots + R_n}{n} \tag{13}$$

4. CASE STUDY

A 5-terminal VSC-MTDC system which connects two OWFs and onshore power grids is used to test the probabilistic stability assessment (PSA) of the hybrid AC/DC system. Two of the GSVSCs are connected to the modified IEEE 39-bus system at bus 2 and bus 9, respectively. Another GSVSC connects an isolated grid which is modified by the WSCC 9-bus system. Each OWF contains 60 DFIGs with a rated power of 4MW. The generator at bus 34 is out of service because of the integration of VSC-MTDC system. The topology of the system is shown in Fig. 4. The test system is developed in DIgSILENT/ PowerFactory. The parameters of DFIG and DC cable are obtained from the library of DIgSILENT/PowerFactory.

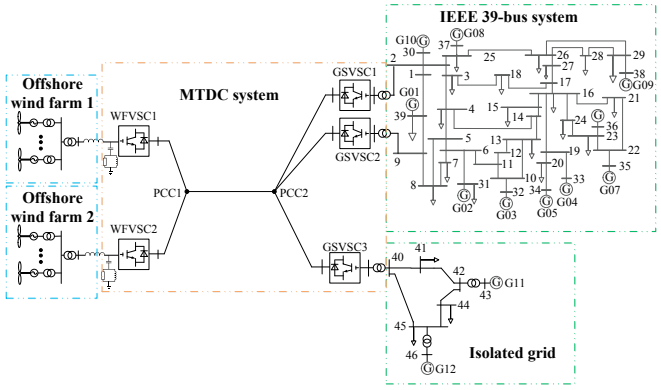


Fig. 4 The diagram of the test system

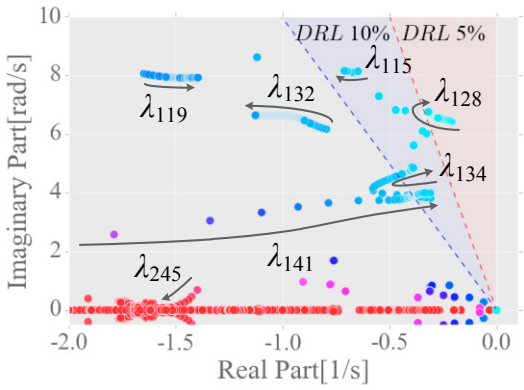
4.1 The influence of the VSG parameters

The most significant parameters of the VSG strategy are H_{vsc} and D. The inertial time constant of synchronous generator is in a range of 2~10 s. However, one of the advantages of the VSG strategy is that the H_{vsc} can be changed by the requirement of the specific system. To investigate the influence of H_{vsc} and D on oscillation damping, the value of H_{vsc} is changed from 0 to 10s with a step of 0.1s and D is changed from 0 to 50 with a step of 1. The root locus of the hybrid system is shown in Fig. 5.

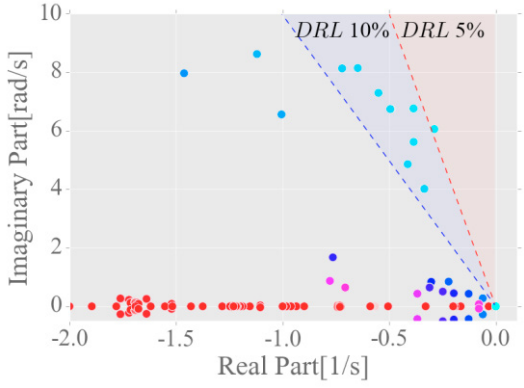
With the increase of H_{vsc}, the damping ratio of some eigenvalues (e.g. λ_{126}, λ_{115}, λ_{132}) increases, while some others decreases (e.g. λ_{141} λ_{119}). Specifically, the traces of λ_{130}, λ_{134}, λ_{128} have a complicated movement. The damping ratio of λ_{134}, λ_{128} increases at first and then decreases from the inflection point. λ_{130} have the opposite performance. Thus, it is important to find a reasonable H_{vsc} to enhance the stability of the hybrid AC/DC system.

The change of D has a little impact on the eigenvalues, which means that D is not a critical parameter for system stability.

To test the impact of different combinations of H_{vsc} and D on the damping oscillation, a three-phase fault at the bus 16 in 39-bus system has been set. After 4 cycles (0.08s), the fault is cleared. The power oscillation of the transmission line 1-2 is shown in Fig. 6.



(a)



(b)

Fig. 5 The root locus: (a) change H_{vsc};(b) change D

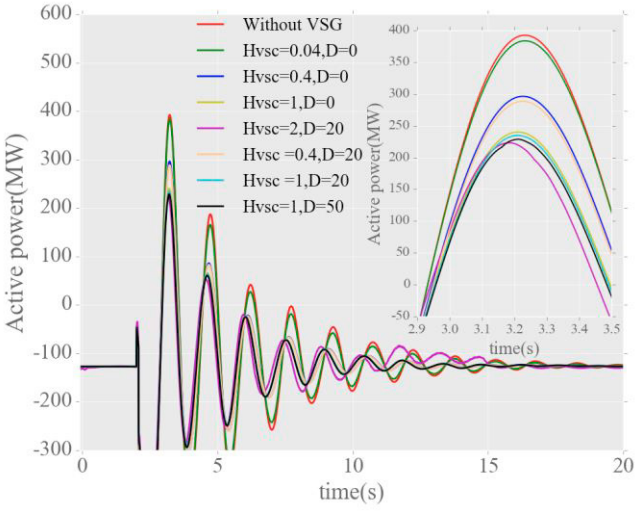


Fig. 6 The power oscillation under different parameters

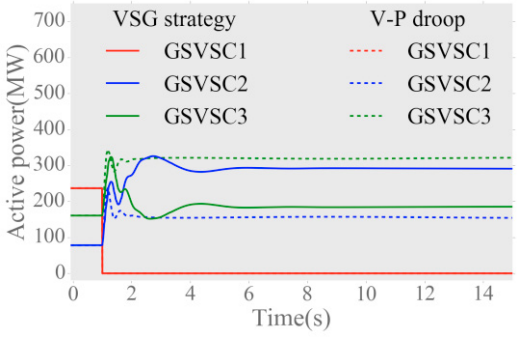
Fig. 6 verifies the result of root locus that H_{vsc} have a significant effect on oscillation damping. However, when H_{vsc} is greater than 1s, there is no more enhancements for the oscillation damping. According to Fig. 5 and Fig. 6, the H_{vsc} and D should be set to 1s and 10.

4.2 The frequency support by VSG strategy

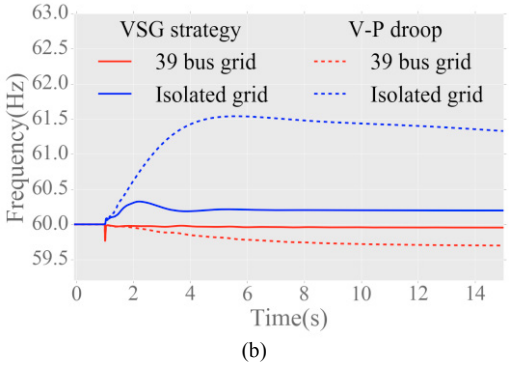
The ability to provide frequency support of VSG strategy can be verified by several cases. Considering the page limit, a GSVSC outage case is used to show the performance of the VSG strategy.

GSVSC1 is out of service at t = 1s and the power output of GSVSC1 (236.41MW) is immediately redistributed to GSVSC2 and GSVSC3. Under the traditional droop strategy, the remaining GSVSCs have no ability to adjust output power. The generators 11 and 12 in the isolated grid decrease the generation power significantly and the rotor speeds increase rapidly. It can be observed from the dashed lines of Fig. 7(b) that the bulk power injection results in a large frequency deviation because of the small size of the isolated grid.

With the VSG strategy, the power ratio between GSVSC2 and GSVSC3 is not fixed at 1:2. The power output of GSVSC is continually adjusted according to the frequency deviation of the adjacent AC system. Thus, much more power is allocated to GSVSC2 while GSVSC3 maintain nearly its initial output power (Fig. 7(a)). Fig. 7(b) shows the effectiveness of reduction of the frequency deviations in two onshore grids.



(a)



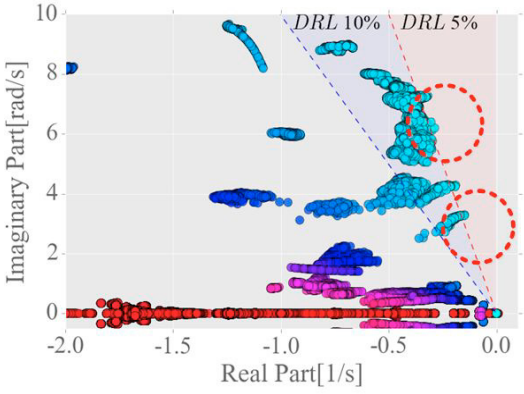
(b)

Fig. 7 System responses in the case of the GSVSC outage: (a) power output of GSVSCs; (b) grid frequency

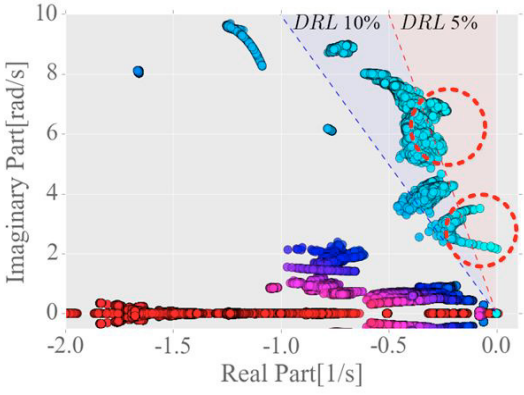
4.3 PSSSA for VSC-MTDC with VSG strategy

PSSSA was performed on a Samsung laptop with Intel (R) Core (TM) i7-5500U, 2.9 GHz processing speed, and 8 GB RAM.

20,000 trials are generated according to the combination of random wind speed, random load demand and random transmission tie-line fault. The eigenvalues of the dominant oscillation modes are shown in the Fig. 8. Fig. 8(b) indicates that more eigenvalues are located in the weak damping area (the red area) when the system adopted the traditional droop method. Fig. 8(a) shows that less eigenvalues located in the weak damping area. The VSG strategy is capable to increase the damping ratio of the poorly damped modes significantly and the instability probability of oscillation modes is eliminated.



(a)



(b)

Fig. 8 The eigenvalues distribution on the s-plane: (a) VSG strategy; (b) traditional droop control method

The comparison of the instable probabilities for the system with different control strategies is shown in Fig. 9.

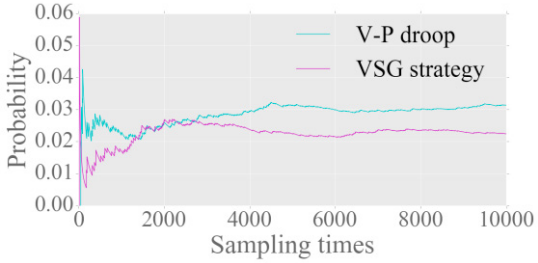


Fig. 9 The convergence of the different control method

Fig. 9 shows that the VSG strategy could enhance the stability of the hybrid AC/DC system. The instability probability of the system with traditional droop method is 3.16% while the VSG strategy is only 2.24%.

5. CONCLUSION

This paper proposed a new virtual synchronous generator strategy for VSC-MTDC system. To minimize the frequency deviation, the DC grid could mimic the action of the prime mover's valve to change the power injected into the GSVSC. Additional damping ability is obtained by the VSG strategy for the VSC-MTDC system to damp the power oscillation. The Monte Carlo method is used to test the stability of the hybrid AC/DC system. The results show that the probabilistic small stability of the hybrid AC/DC system which adopts VSG strategy is enhanced significantly.

ACKNOWLEDGMNETS

This work was supported by the project on "Advanced Application and Simulation System for Low Frequency Oscillation Analysis" under Grant GDDW2120160303-AS00187.

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